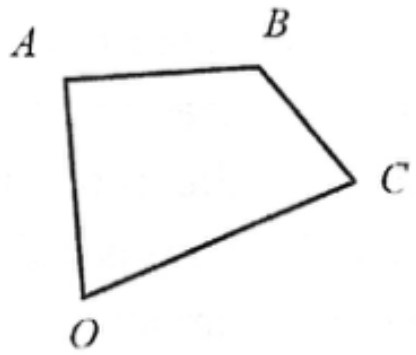
2D Vectors

Questions 1–23

Question 1. Given $\mathbf{a} = 2\mathbf{i}+\mathbf{j}$, $\mathbf{b} = 3\mathbf{j}$ and $\mathbf{c} = -\mathbf{i}-4\mathbf{j}$ then find the following, giving your answer both in component and column vector form:

a) $3\mathbf{a}$
b) $2\mathbf{b}+\mathbf{c}$
c) $\mathbf{a}-2\mathbf{c}$

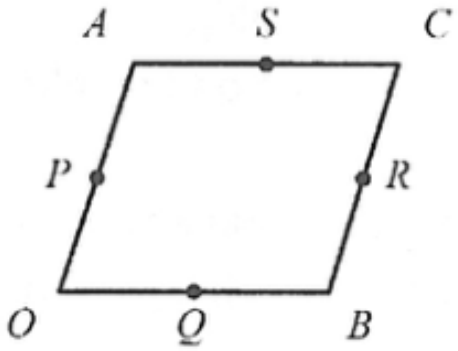
Question 2. For the quadrilateral shown, $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{OC} = \mathbf{c}$



find the following in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$:

a) $\overrightarrow{AB}$
b) $\overrightarrow{CB}$
c) $\overrightarrow{AC}$

Question 3. For the parallelogram shown, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$



The midpoints of OA, OB, BC and AC are P, Q, R and S respectively. Find in terms of $\mathbf{a}$ and $\mathbf{b}$ each of the following:

a) $\overrightarrow{OP}$
b) $\overrightarrow{OQ}$
c) $\overrightarrow{OR}$
d) $\overrightarrow{OS}$
e) $\overrightarrow{BQ}$
f) $\overrightarrow{SR}$
g) $\overrightarrow{RP}$
h) $\overrightarrow{AQ}$

Question 4. Given two non-parallel vectors $\mathbf{a}$ and $\mathbf{b}$, state whether each of the following pairs of vectors are parallel:

a) $4\mathbf{a}$ and $-2\mathbf{a}$
b) $2\mathbf{a} + \mathbf{b}$ and $\mathbf{a} + \frac{1}{2}\mathbf{b}$
c) $3\mathbf{a} - \mathbf{b}$ and $-3\mathbf{a} - \mathbf{b}$
d) $-2\mathbf{a} - 4\mathbf{b}$ and $8\mathbf{a} + 24\mathbf{b}$

Question 5. Find the magnitude of the following vectors:

a) $\mathbf{a} = 4\mathbf{i} + 5\mathbf{j}$

b) $\mathbf{b} = \begin{pmatrix} 7 \\ 1 \end{pmatrix}$

c) $\mathbf{c} = -3\mathbf{i} + \mathbf{j}$

d) $\mathbf{d} = \begin{pmatrix} p \\ q \end{pmatrix}$

Question 6. Find the magnitude and the bearing of each of the following vectors, where vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors due east and north respectively:

a) $\mathbf{a} = 2\mathbf{i} + 12\mathbf{j}$

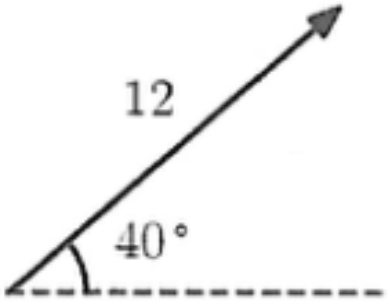
b) $\mathbf{b} = -5\mathbf{i} + 3\mathbf{j}$

c) $\mathbf{c} = -10\mathbf{i} - 5\mathbf{j}$

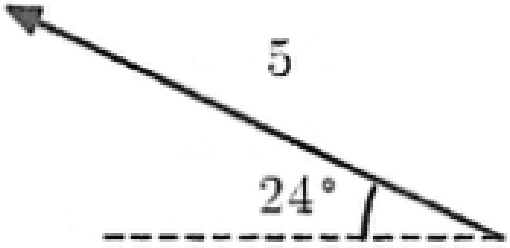
d) $\mathbf{d} = 8\mathbf{i} - 15\mathbf{j}$

Question 7. Write the following in the form $p\mathbf{i} + q\mathbf{j}$, giving p and q to 3 significant figures:

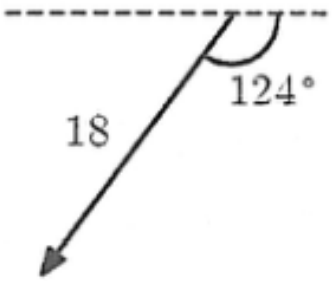
a)



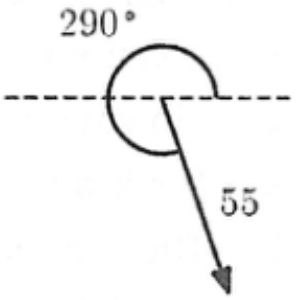
b)



c)



d)



Question 8. Find the unit vector in the direction of the following vectors:

a) $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$

b) $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$

c) $\mathbf{c} = 5\mathbf{i} - 8\mathbf{j}$

Question 9. The vector $\mathbf{a}$ is parallel to the given vector $\mathbf{p}$. In each case, given the magnitude of $\mathbf{a}$, find the vector $\mathbf{a}$:

a) $\mathbf{p} = -2\mathbf{i} + 6\mathbf{j}$ and $|\mathbf{a}| = 3\sqrt{10}$
b) $\mathbf{p} = 2\mathbf{i} + 7\mathbf{j}$ and $|\mathbf{a}| = \frac{1}{2}\sqrt{53}$
c) $\mathbf{p} = \mathbf{i} - 8\mathbf{j}$ and $|\mathbf{a}| = 3\sqrt{65}$

Question 10. State the position vector of each of the following coordinates:

a) $(1, 2)$
b) $(-1, 6)$
c) (p, q)

Question 11. It is given that A, B and C have the coordinates $(1, 2)$, $(-2, 6)$ and $(3, -10)$ respectively. Find

a) $\overrightarrow{AB}$

b) $\overrightarrow{BC}$

c) $\overrightarrow{AC}$

d) $\overrightarrow{BA}$

Question 12. Given each of the following position vectors, find the magnitude of the vector $\overrightarrow{AB}$:

a) $\overrightarrow{OA} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$

b) $\overrightarrow{OA} = 2\mathbf{i} - 7\mathbf{j}$ and $\overrightarrow{OB} = 9\mathbf{i} + 4\mathbf{j}$

Question 13. Given each of the following position vectors, find the unit vector in the direction of the vector $\overrightarrow{AB}$:

a) $\overrightarrow{OA} = \begin{pmatrix} 1 \\ -5 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$

b) $\overrightarrow{OA} = 5\mathbf{i} + 2\mathbf{j}$ and $\overrightarrow{OB} = -3\mathbf{i} + 5\mathbf{j}$

Question 14. Find the angle between each of the following pair of vectors, giving your answer to 1 decimal place:

a) $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 5 \\ 0 \end{pmatrix}$

b) $2\mathbf{i} + 7\mathbf{j}$ and $3\mathbf{i} + 3\mathbf{j}$

c) $2\mathbf{i} - 10\mathbf{j}$ and $8\mathbf{i} - 3\mathbf{j}$

d) $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$

e) $\mathbf{i} - 2\mathbf{j}$ and $-\mathbf{i} + 7\mathbf{j}$

Question 15. It is given that $\overrightarrow{AB} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and the angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$ is θ. Show that $\cos\theta = -\frac{\sqrt{2}}{2}$.

Question 16. It is given that $\overrightarrow{OA} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 4 \\ 8 \end{pmatrix}$ and $\overrightarrow{OC} = \begin{pmatrix} -5 \\ 1 \end{pmatrix}$. It is known that $\overrightarrow{BC} = \overrightarrow{AD}$, find the position vector of the point D.

Question 17. In each case, determine whether each set of points A, B and C with the given coordinates are respectively collinear:

a) $(-8, 2)$, $(0, 7)$ and $(8, 12)$
b) $(-5, 4)$, $(4, -5)$ and $(7, -8)$
c) $(-10, 17)$, $(-3, 13)$ and $(5, 8)$

Question 18. The midpoint of A and B is M. Point A has position vector $-5\mathbf{i} + 13\mathbf{j}$ and the position vector of M is $-\mathbf{i} + 4\mathbf{j}$. Find the position vector of the point B.

Question 19. The position vectors of A, B and C respectively are $\mathbf{a} = \begin{pmatrix} -3 \\ 7 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} 1 \\ k \end{pmatrix}$. There is a further point D $(p, 8)$ such that $\overrightarrow{AB} = \overrightarrow{CD}$.

a) Find the value of p and k

b) Determine the shape of $ABDC$.

Question 20. The points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. A third point C forms the parallelogram $OABC$.

a) In terms of $\mathbf{a}$ and $\mathbf{b}$,

i) find $\overrightarrow{OC}$

ii) find $\overrightarrow{OP}$, where P is the midpoint of AB

iii) find $\overrightarrow{OQ}$, where Q is the midpoint of OC

b) Show that the midpoint of $\overrightarrow{PQ}$ is at the centre of the parallelogram.

Question 21. A quadrilateral is given by $ABCD$. It is known that $\overrightarrow{AB} = 4\mathbf{i} + 5\mathbf{j}$ and $\overrightarrow{DC}$ is the same length and direction as $\overrightarrow{AB}$. Two points C and D have position vectors $\mathbf{c} = 5\mathbf{i} + \mathbf{j}$ and $\mathbf{d} = k\mathbf{i} - 4\mathbf{j}$ respectively. Find

a) the value of k

b) the coordinates of A and B given $ABCD$ is a square.

Question 22. The position vectors of A, B and C respectively are $\mathbf{a} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -5 \\ 4 \end{pmatrix}$. The midpoint of A and B is P and the midpoint of B and C is Q.

a) Find the coordinates of the points P and Q

b) Show that $\overrightarrow{PQ} = \frac{1}{2}\overrightarrow{AC}$

A further point D exists such that $|\overrightarrow{DB}| = \sqrt{74}$ and $\overrightarrow{BC}$ is parallel to $\overrightarrow{AD}$

c) Find the possible coordinates of D

Question 23. The points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ relative to the origin O where $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 10 \end{pmatrix}$. Find the possible coordinates of a third point C such that $\overrightarrow{AB}$ is the hypotenuse of a right-angled isosceles triangle.